

RQ1: Profile accuracy and abstention

Research question

RQ1. Can the pipeline construct accurate site profiles, and does it abstain where a site is silent rather than guessing?

The analysis compares the final evidence-backed profiles against hand-built ground truth at three sites. It evaluates populated values, correct and false abstentions, unsupported values, evidence conflicts, and citation integrity.

Run configuration

- **60 completed profile builds:** 20 each for Anvil, Stampede3, and ND CRC.
- **Models:** GPT-5 Mini, GPT-5.6 Terra, Gemini Flash, and Gemini Pro.
- **Retrieval:** one BM25 run, three LLM-expanded BM25 runs, and one full-corpus run per site-model pair.
- **Frozen inputs:** every run for a site uses the same login measurement, pilot result, and documentation corpus. No evaluation run performs new discovery, measurement, or pilot submission.
- **Ground truth:** 147 fields for Anvil, 159 for Stampede3, and 68 for ND CRC, producing 7,480 run-field observations.
- **Balancing:** BM25 and full corpus have weight one; each expanded repetition has weight one third. Every model-retrieval configuration therefore has equal aggregate weight.
- **Matching:** populated values use field-specific numeric, set, pattern, or normalized-text rules. Site silence and not-applicable states are both evaluated as expected absence in this preliminary analysis.

The analysis is reproducible with:

```
python evaluation/rq1/run_rq1_analysis.py
```

Field outcomes

Site	Precision	Recall	Abstention accuracy	Field accuracy
Anvil	0.977	0.867	0.969	0.889
Stampede3	0.957	0.906	0.935	0.912
ND CRC	0.990	0.764	0.972	0.777
Macro mean	0.975	0.846	0.959	0.859

Precision exceeds recall in the macro result and at all three sites. The predicted abstention rate is highest at Anvil (0.306), followed by ND CRC (0.274) and Stampede3 (0.238).

Configuration fields are consistently strongest. Policy accuracy is lower: 0.821 at Anvil, 0.876 at Stampede3, and 0.809 at ND CRC. This is the portion of the profile most dependent on documentation extraction.

Evidence behavior

The profiles expose configuration-versus-policy disagreement. Stampede3 produces six walltime conflicts per balanced configuration. Anvil produces 7.33 of its eight expected walltime conflicts on average; one configuration failed to recover the documented limits. Anvil also contains 0.28 other conflicts per configuration, primarily an erroneous GPU-count disagreement in a few runs.

GPU facts demonstrate both recovery and difficulty. Stampede3 recovers the H100 and PVC counts and models in every configuration. Anvil recovers the A100 model for the `ai` and `gpu` partitions in 0.222 of configurations and for `gpu-debug` in 0.028.

Citations

All 3,052 accepted citations are mechanically valid: the cited chunk exists, belongs to the target site, uses the frozen source URL, and contains the quoted text. The 2,257 accepted findings all contain at least one citation. This does not establish semantic entailment; that requires a separate manual audit.

Answer to RQ1

The pipeline constructs mostly accurate and conservative profiles for the three evaluated sites. Macro precision is 0.975, recall is 0.846, abstention accuracy is 0.959, and field accuracy is 0.859. Precision exceeds recall at every site, so the dominant failure is leaving a known value unresolved rather than populating an unsupported value.

The evidence paths are complementary. Login measurement directly establishes 30--38% of evaluated fields per site, documentation recovers another 22--39%, and pilot evidence contributes 4--10%. Correct abstention accounts for 6--21%, leaving 9% unresolved or wrong at Stampede3, 11% at Anvil, and 22% at ND CRC.

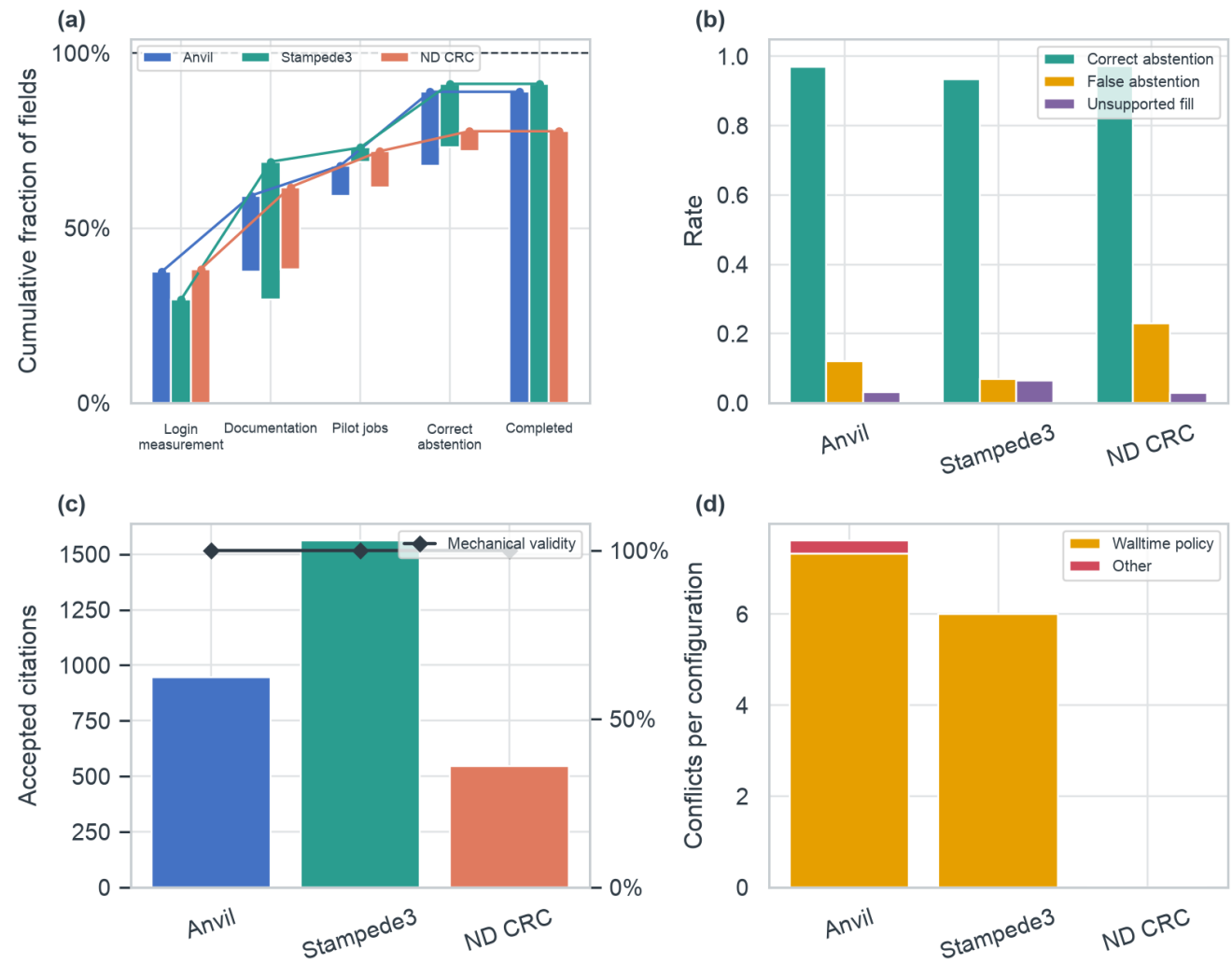
The result is site-dependent. Stampede3 reaches 0.912 field accuracy, Anvil 0.889, and ND CRC 0.777. ND CRC's lower recall shows that abstention is not always correct: some documented values remain unresolved. At the same time, the pipeline identifies measurement-documentation disagreements, including the Slurm walltime conflicts, and recovers GPU facts that login-node measurement misses. RQ1 is therefore answered positively for accuracy and conservative behavior, but not as complete coverage: profiles remain partial where discovery, extraction, or evidence is insufficient.

Figures

All figures use balanced configuration weights: BM25 and full corpus have weight one, while each of the three expanded-retrieval repetitions has weight one third.

The palette uses medium-saturation blue and teal for correct outcomes, red and amber for errors, and a colorblind-distinguishable blue, teal, and coral site encoding. It is designed for clear reproduction in an academic paper.

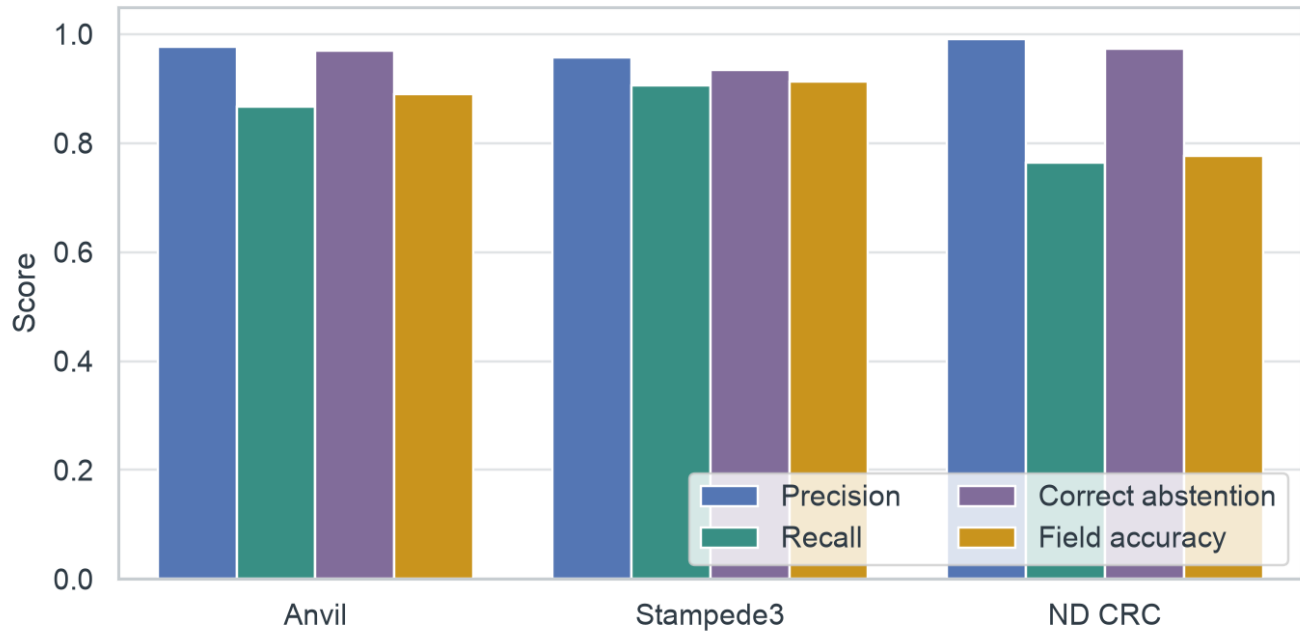
Compact RQ1 summary



PDF

Panel (a) cumulatively shows how login measurement, documentation, pilot evidence, and correct abstention contribute to each completed profile; the gap to 100% is wrong or unresolved. Panel (b) separates correct abstention, false abstention, and unsupported fills. Panel (c) shows accepted citation volume and mechanical validity. Panel (d) shows explicit evidence conflicts, separating walltime-policy disagreements from other conflicts. Precision, recall, and field accuracy are omitted because they are reported in the table above.

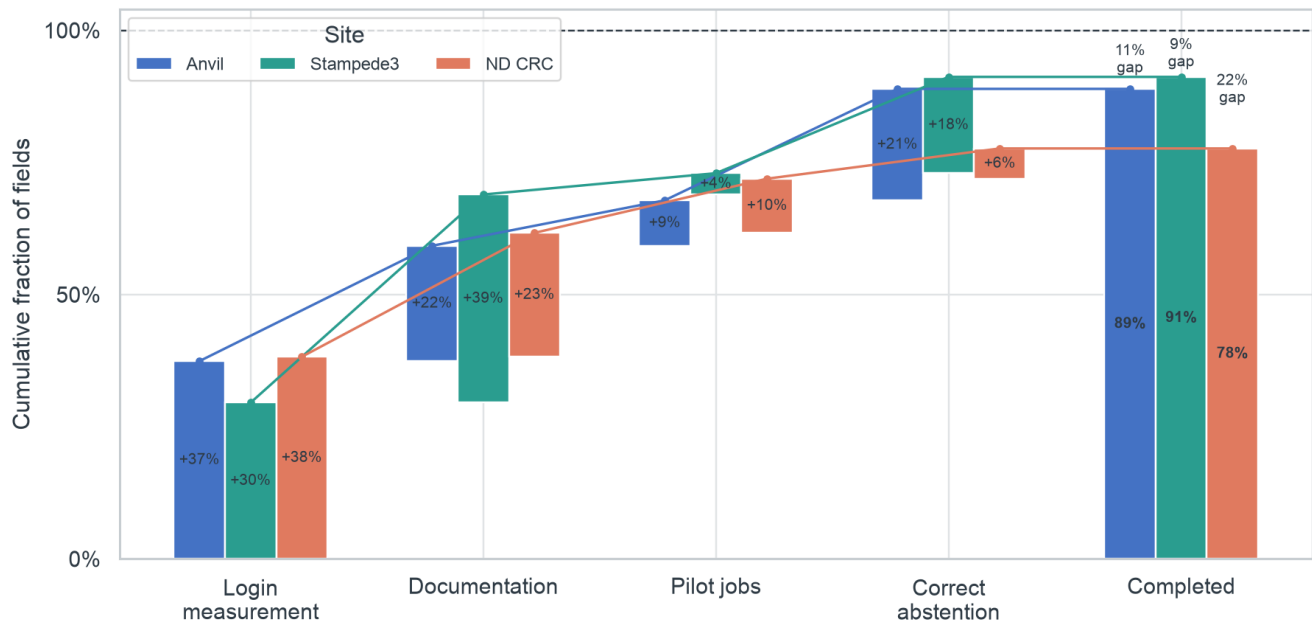
Site metrics



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This grouped comparison separates precision, recall, abstention accuracy, and overall field accuracy by site. ND CRC has the highest precision but the lowest recall because it leaves more ground-truth values unresolved.

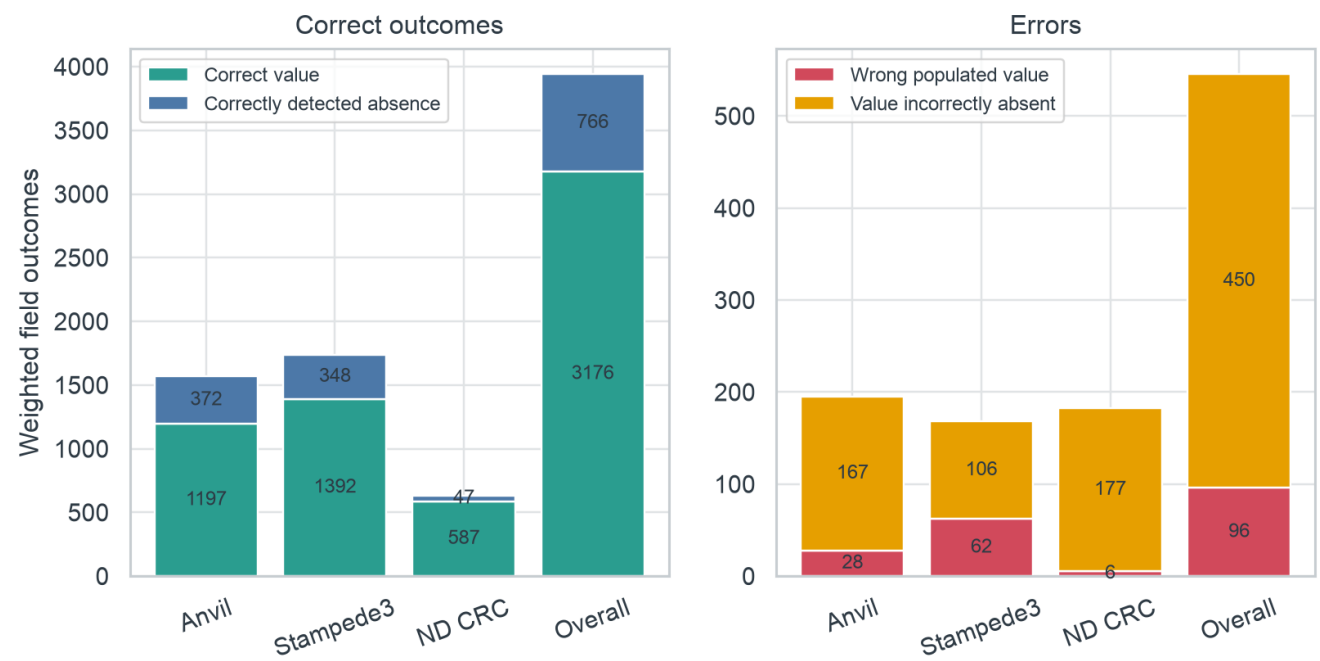
Profile completion waterfall



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Each floating bar adds the fraction of fields established by login measurement, documentation, pilot evidence, or correct abstention. The final bar reports their cumulative completion, while the labeled gap to 100% combines wrong values with unresolved known values.

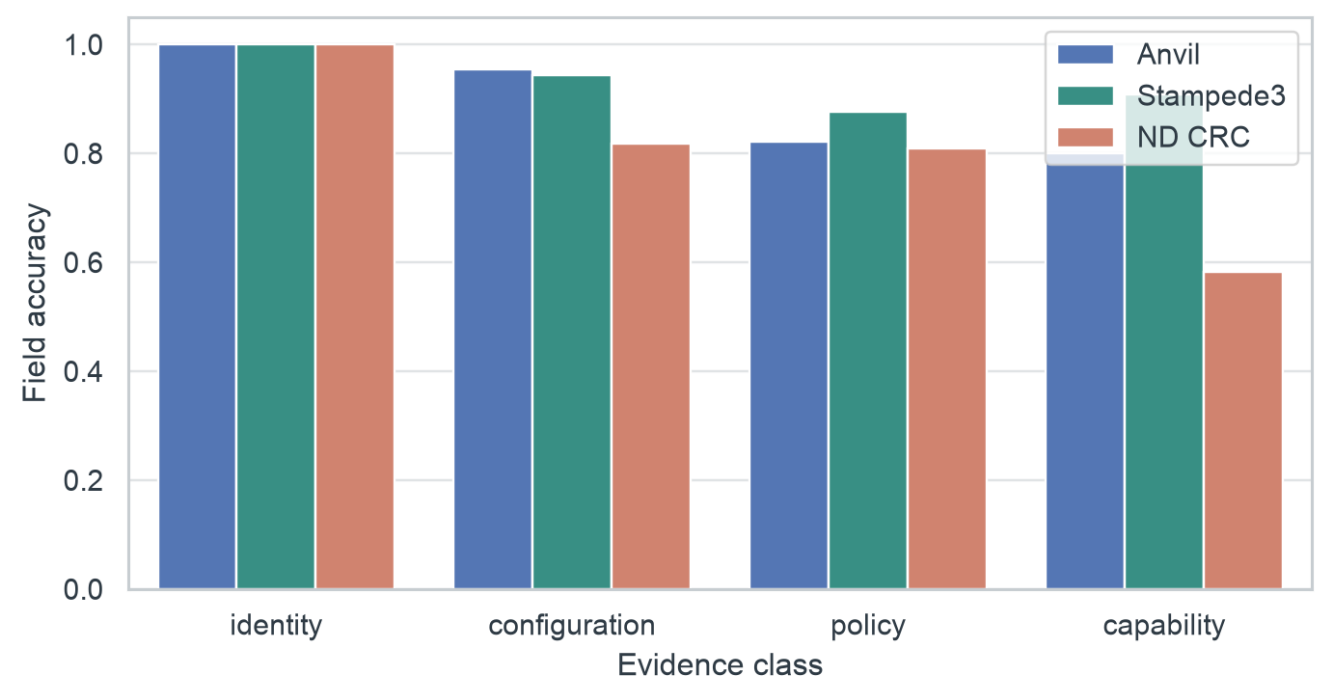
Combined outcome counts



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The left panel gives weighted counts of correct values and correctly detected absences. The right panel gives wrong populated values and values that should have been populated but were absent.

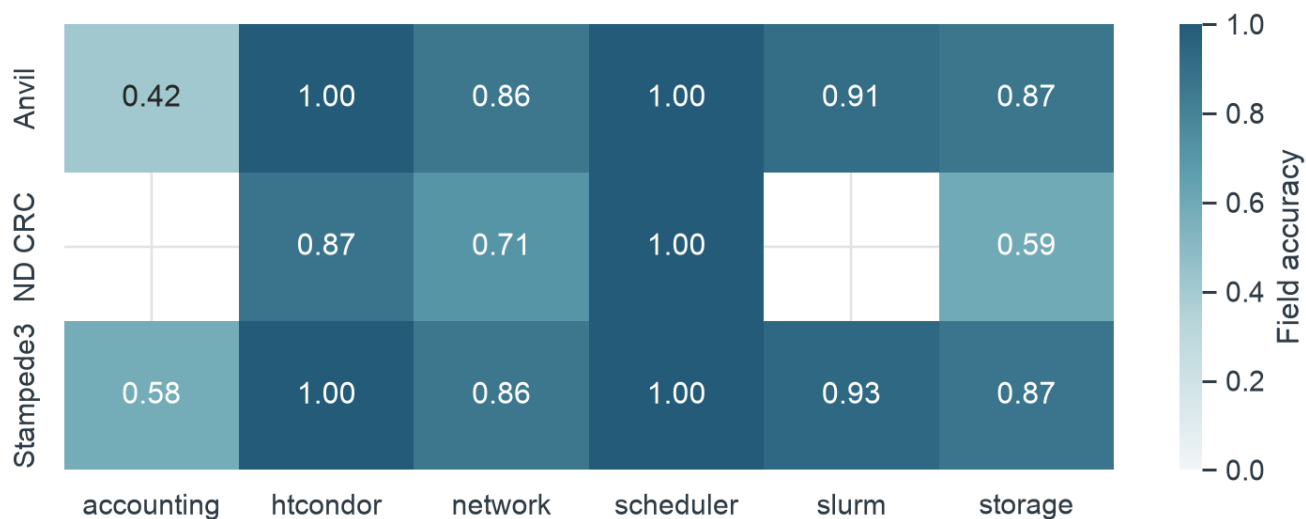
Accuracy by evidence class



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This figure separates identity, configuration, policy, and capability fields. It shows where the full pipeline is strongest and where documentation extraction or pilot coverage still limits accuracy.

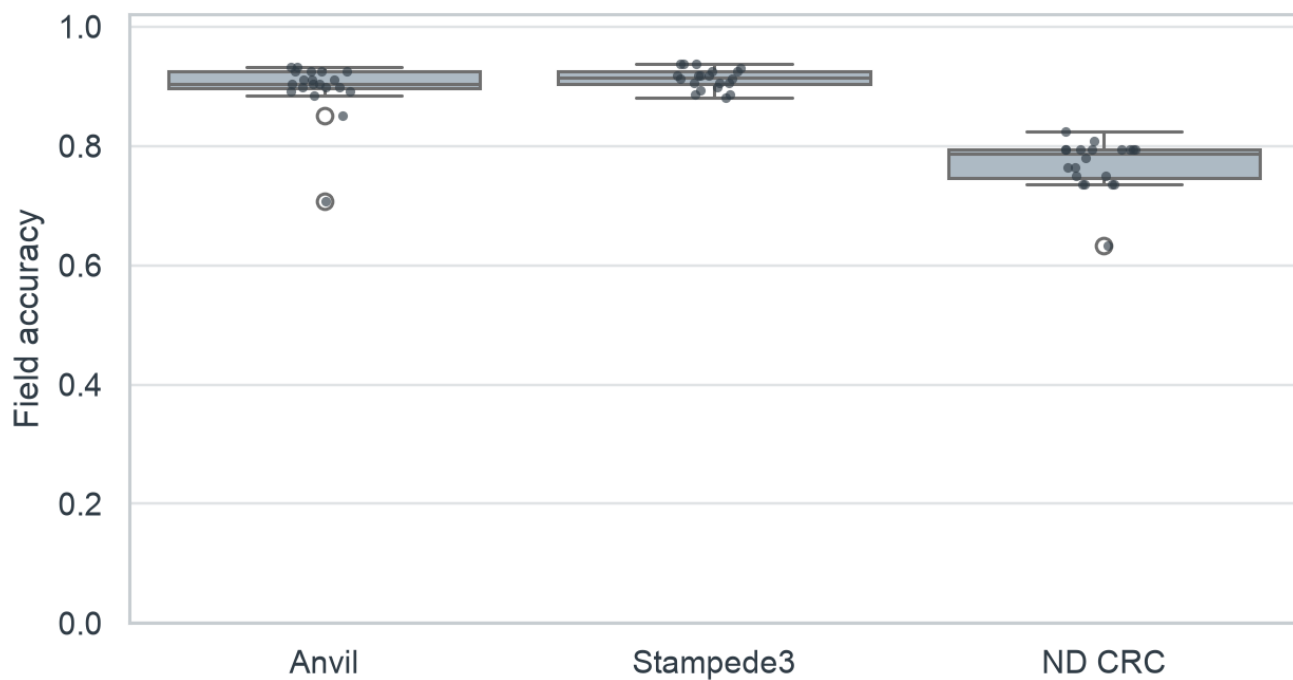
Accuracy by profile section



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The heatmap localizes accuracy by site and profile section, making weaknesses in scheduler, storage, networking, or accounting fields visible without averaging them into one site-level score.

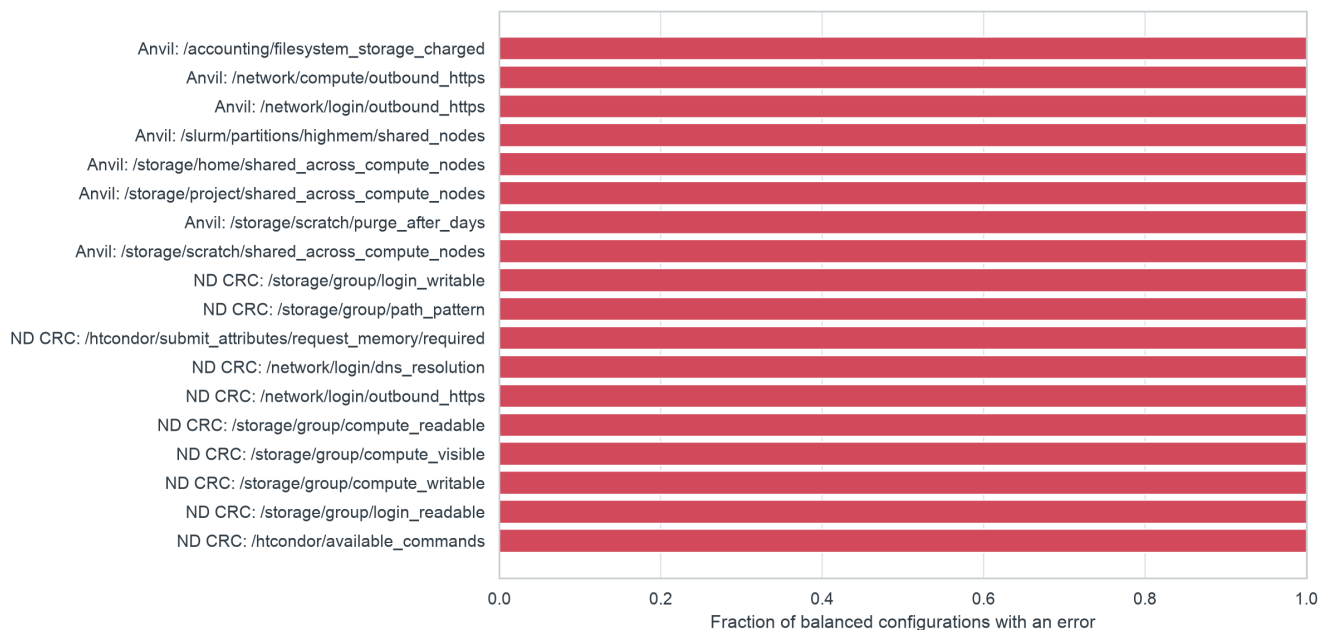
Run accuracy distribution



[PDF](#)

Each point is one model-and-retrieval run; the box summarizes the distribution for a site. The spread measures output variance that is hidden by aggregate precision and recall.

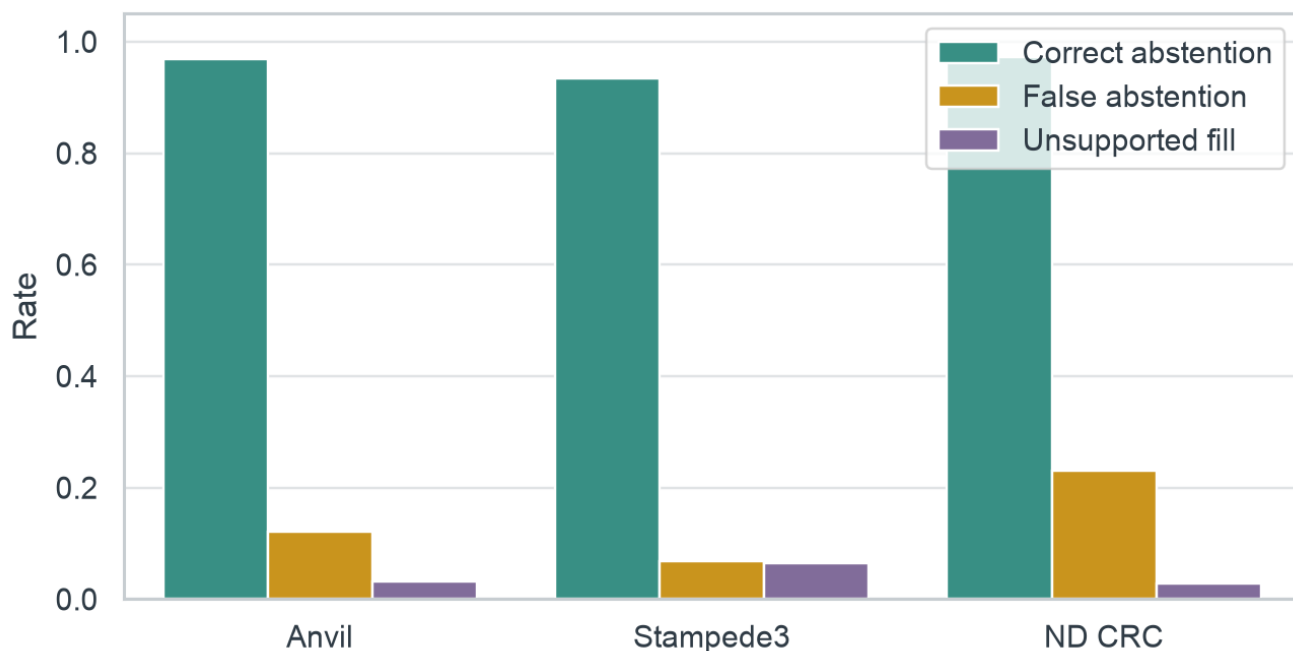
Most frequent field errors



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These are the 18 fields with the highest balanced error rate. An error includes a wrong value, a false abstention, or a value populated where ground truth says the field is absent.

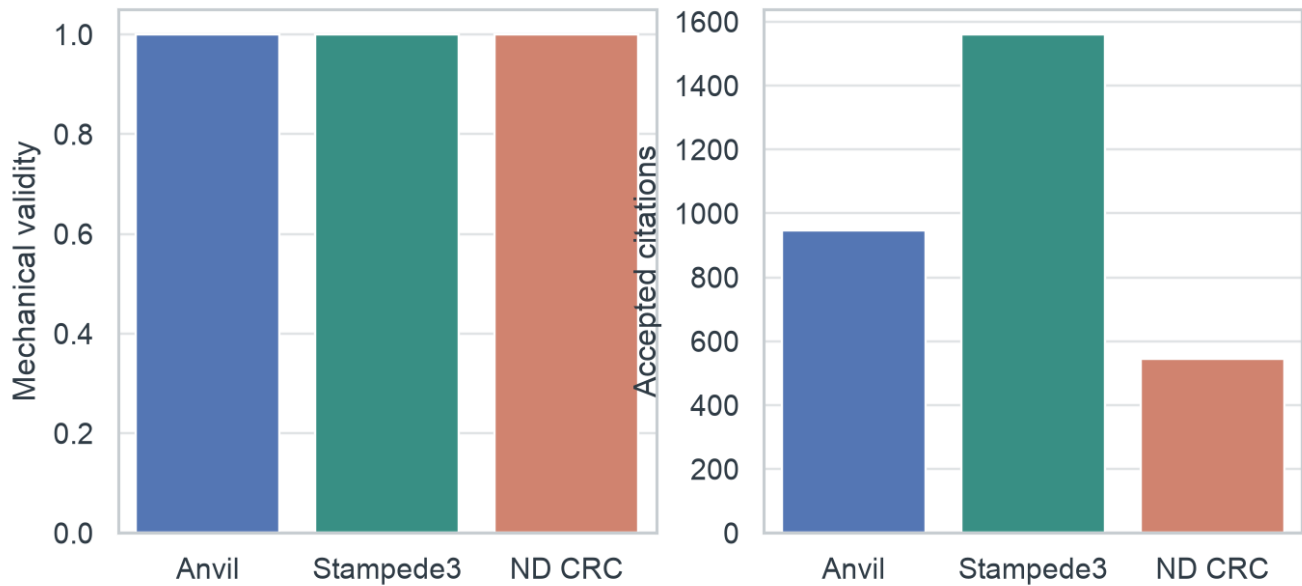
Abstention behavior



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Correct abstention measures silence on truly absent fields. False abstention measures missed known values, while unsupported fill measures values proposed for absent fields. These rates use different denominators and should not be read as parts of one stacked total.

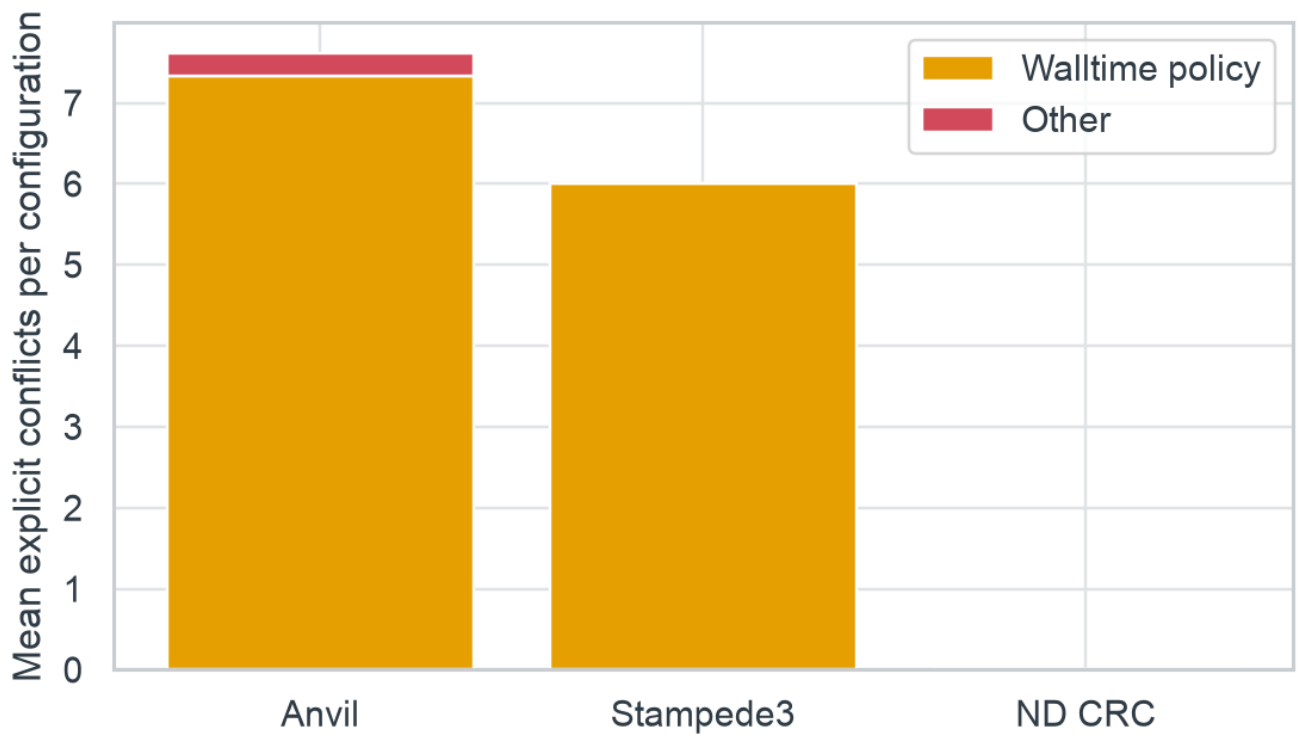
Citation integrity



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The left panel checks that citations point to stored target-site chunks and contain the recorded quote. The right panel shows citation volume. Mechanical validity does not by itself prove that the quote semantically entails the field.

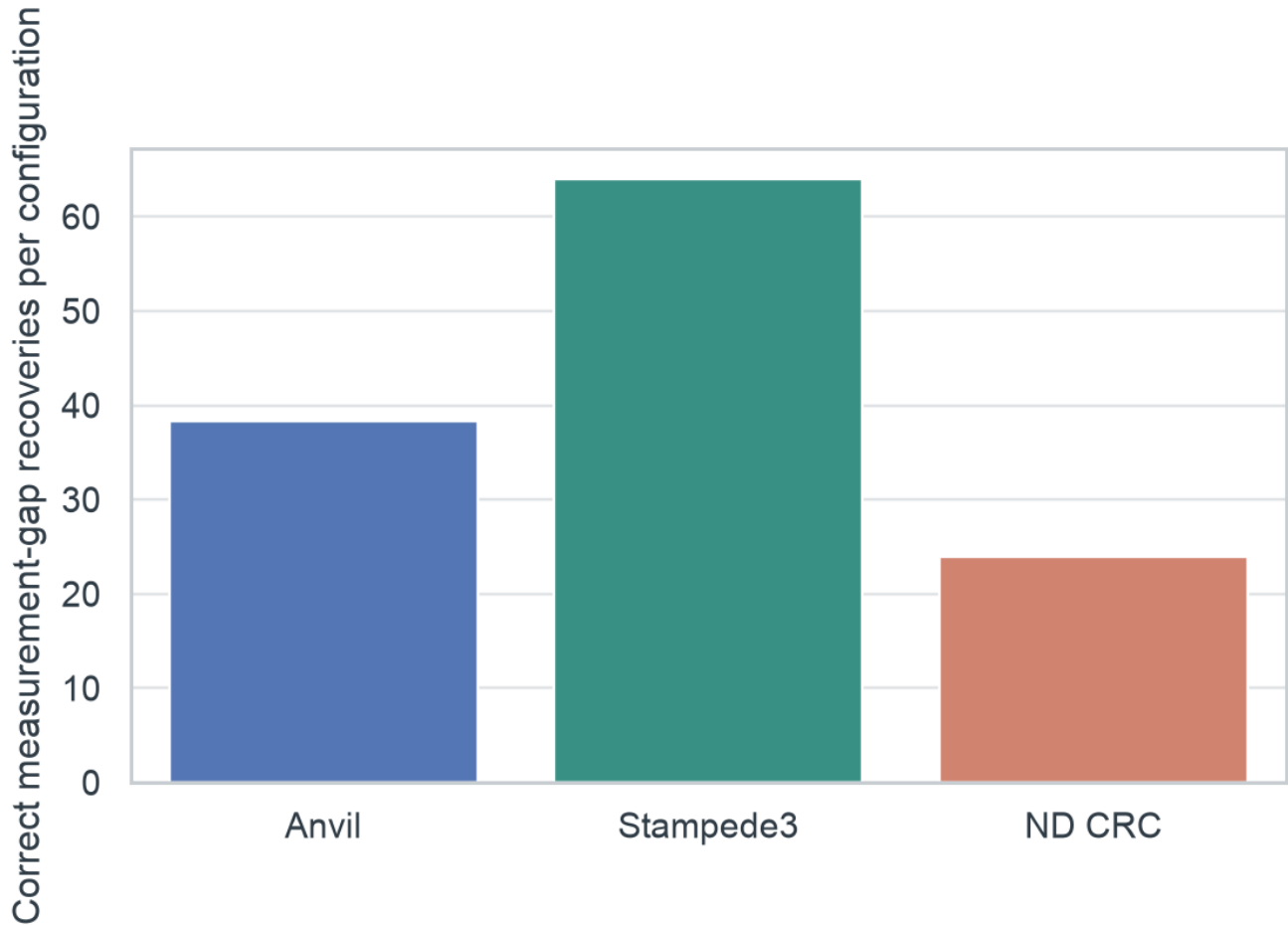
Source conflicts



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This figure counts explicit evidence disagreements per balanced configuration. Walltime conflicts are separated because they capture the central case where visible scheduler configuration reports unlimited time but documentation states an enforced policy limit.

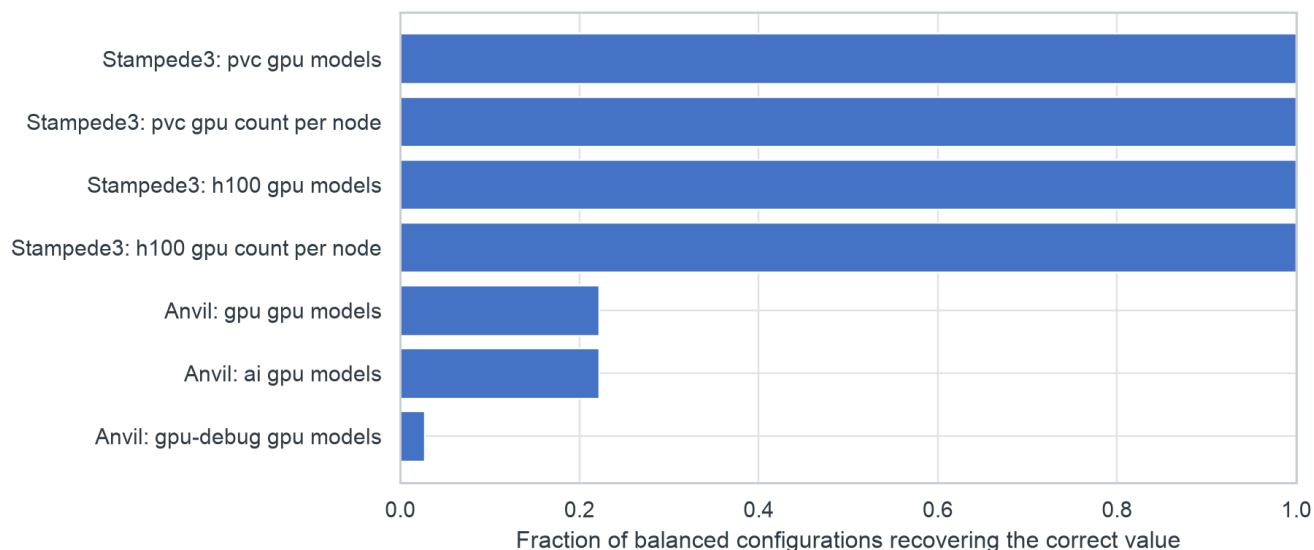
Measurement-gap recovery



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The bars count correct values recovered per configuration when login measurement supplied no value. They quantify what documentation extraction and pilot evidence add beyond measurement alone.

GPU measurement-gap recovery



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This field-level view reports how often the pipeline recovers the correct GPU count or model when login-node measurement cannot see it. Stampede3 is consistently recovered, while the Anvil GPU-model fields remain difficult.

Interpretation boundary

These numbers establish what the current evaluator measures. Claims about semantic citation validity and why a model made an error require manual review. Model and retrieval comparisons belong to RQ2 and are not inferred from the site-level RQ1 aggregates.

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RQ2: Model and retrieval effects

Research question

RQ2. Does the choice of model or retrieval strategy affect what ends up in the profile?

The analysis compares final profile quality after deterministic validation, alongside extraction-token cost, latency, rejected findings, unresolved fields, and run-to-run variance.

Run configuration

- **60 completed profile builds:** 3 sites × 4 models × 3 retrieval modes, with additional repetitions for the stochastic expanded mode.
- **Sites:** Purdue Anvil, TACC Stampede3, and Notre Dame CRC.
- **Models:** GPT-5 Mini, GPT-5.6 Terra, Gemini Flash, and Gemini Pro.
- **Retrieval:** BM25, LLM-expanded BM25, and full corpus.

- **Repetitions:** one BM25 and one full-corpus run per site-model pair; three LLM-expanded BM25 runs per site-model pair.
- **Frozen inputs:** every run for a site uses the same login measurement, pilot result, and documentation corpus. No run performs new discovery, measurement, or pilot submission.
- **Balancing:** BM25 and full corpus have weight one; each expanded repetition has weight one third. Thus every site-model-retrieval configuration has equal aggregate weight.
- **Cost boundary:** extraction tokens and profile-build latency are compared here. One-time corpus-discovery tokens are excluded because the corpus is frozen and shared by every run at a site.

Results by model

Model	Field accuracy	Precision	Recall	Tokens	Latency (s)	Rejected	Unresolved
GPT-5 Mini	0.826	0.953	0.810	83,951	298.1	19.4	44.3
GPT-5.6 Terra	0.866	0.982	0.851	56,793	48.2	8.2	42.9
Gemini Flash	0.876	0.982	0.865	82,454	141.8	6.2	41.6
Gemini Pro	0.869	0.981	0.857	84,059	200.9	13.0	41.7

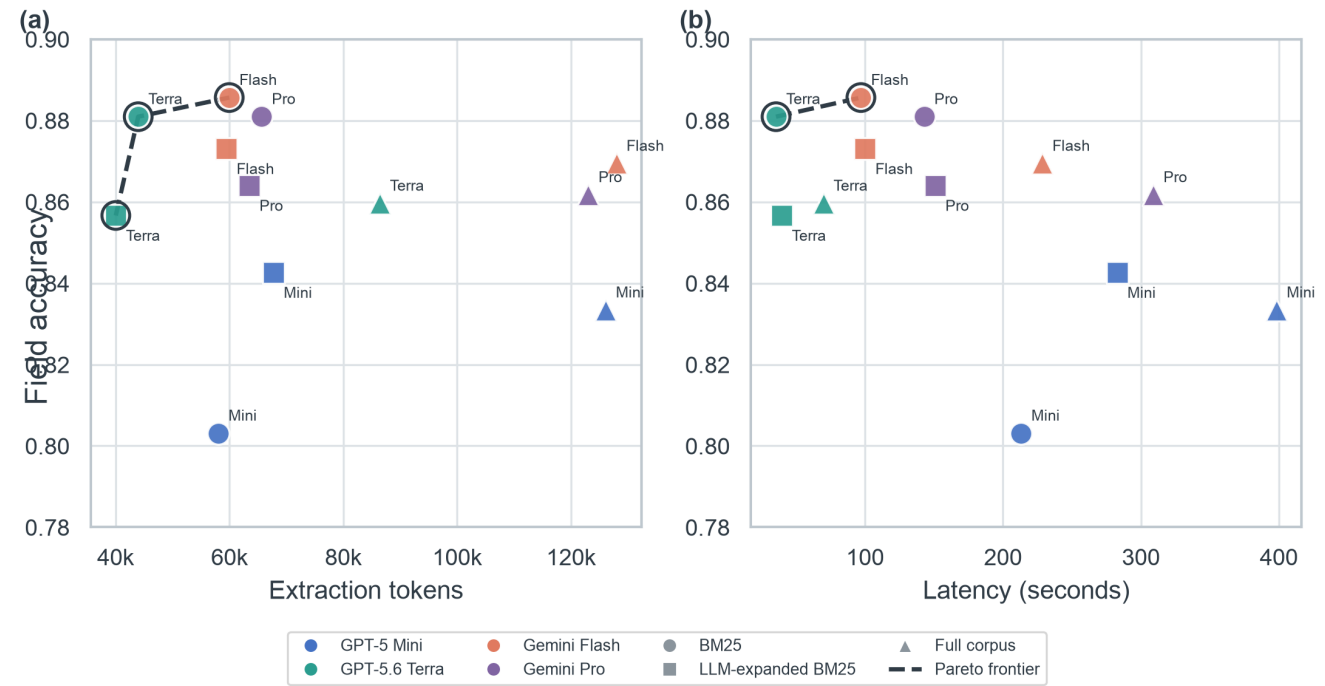
The model-level field-accuracy spread is 0.050 (5.0%). Gemini Flash has the highest balanced field accuracy. GPT-5.6 Terra is within 0.010 (1.0%) of that result while using 31.1% fewer extraction tokens and 66.0% less time than Gemini Flash.

Results by retrieval strategy

Retrieval	Field accuracy	Precision	Recall	Tokens	Latency (s)	Model calls	Rejected	Unresolved
BM25	0.863	0.972	0.849	56,900	122.1	4.7	5.0	42.1
LLM-expanded BM25	0.859	0.975	0.847	57,678	143.3	5.9	13.8	42.1
Full corpus	0.856	0.976	0.842	115,864	251.4	13.8	16.3	43.7

The retrieval-level field-accuracy spread is only 0.007 (0.7%). Relative to BM25, full corpus uses 2.04× as many extraction tokens and takes 2.06× as long without improving aggregate field accuracy. LLM-expanded BM25 uses 1.01× the tokens and 1.17× the time of BM25, also without an aggregate accuracy improvement.

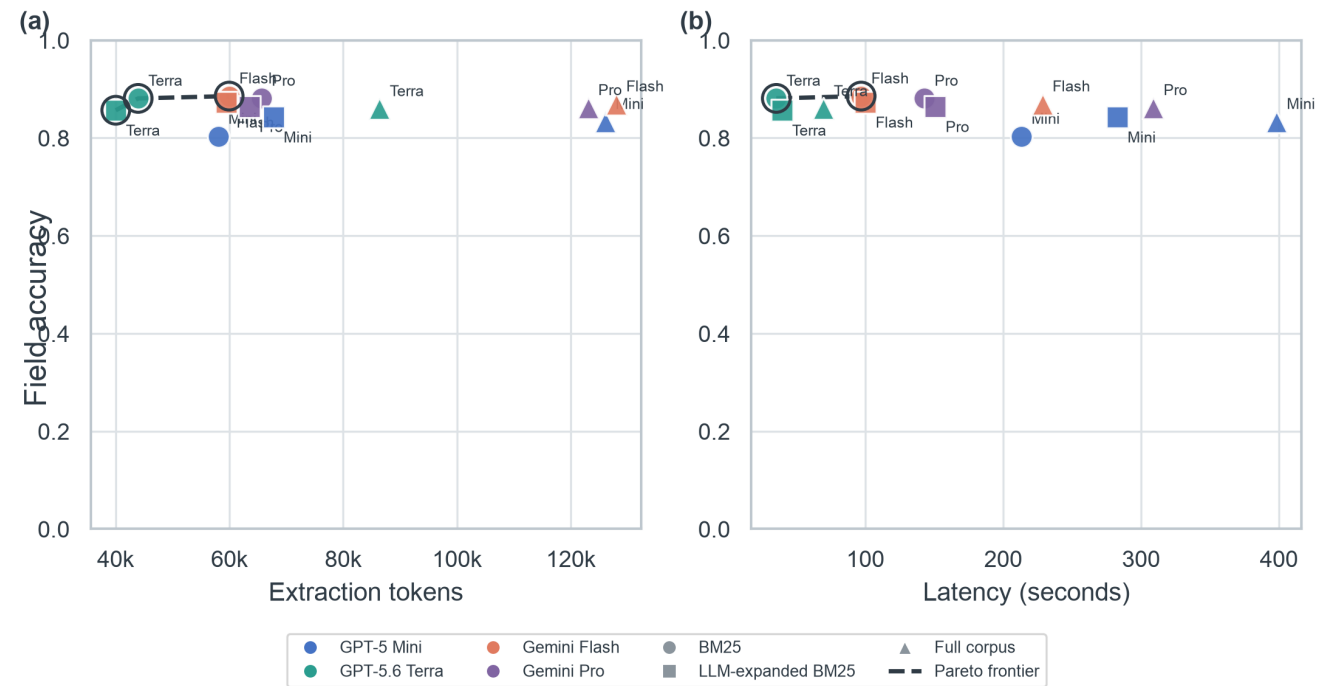
Quality-cost Pareto comparison



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Panel (a) compares field accuracy against extraction tokens; panel (b) compares accuracy against latency. Color identifies the model, point shape identifies the retrieval strategy, and all markers have the same size. The dashed line and outlined points mark the Pareto frontier: configurations for which no other point is both cheaper and at least as accurate. **The field-accuracy axis is truncated to expose differences among configurations.**

Full-scale Pareto reference



Vector PDF figure

This version uses the full 0–1 field-accuracy scale so the absolute magnitude of the differences remains visible.

Run-to-run variance

Across expanded-mode site-model groups, the mean field-accuracy standard deviation is 0.020. The largest is 0.075 for GPT-5 Mini on ND CRC. This variance is concentrated in particular site-model combinations rather than uniformly distributed.

Answer to RQ2

Retrieval strategy has little effect on aggregate profile quality in this experiment, but a large effect on cost: full corpus roughly doubles both tokens and latency. Model choice has a larger quality effect than retrieval choice, and an even larger latency effect. GPT-5.6 Terra provides the strongest quality-cost balance in this matrix: its accuracy is close to the best observed model while its token use and latency are substantially lower.

These are descriptive results over three sites. The deterministic validator constrains accepted outputs, but it does not make all models identical: individual hard fields and some site-model combinations still diverge.